Audio Routing & Conference Mode

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GCXONE supports advanced audio routing logic that allows multiple operators and field devices to collaborate in real-time. This is primarily powered by a Twilio Conference Bridge, ensuring high-quality, low-latency duplex communication.

Audio Routing Modes

Routing behavior is determined by the Active status of the Audio Devices configured at any given site.

Conference Mode (Full Collaboration)

Condition: All configured audio devices at the site are toggled to Active.

Behavior: All active devices and the operator are connected via a central conference bridge.

Best For: Command centers or guard booths with multiple intercoms and speakers that need to hear the same broadcast simultaneously.

Dial-In Mode (Direct Control)

Condition: Exactly one audio device is Active — all others are Inactive.

Behavior: The operator is linked directly to the single active device. All other hardware is ignored.

Best For: Single-point office intercoms or dedicated phone-line integrations.

Device Audio Mode (Isolated)

Condition: All audio devices at the site are Inactive.

Behavior: The system uses the camera's default internal microphone and speaker via G.711 or AAC over RTSP.

Best For: Basic cameras with built-in audio support but no external speaker system.

Configuration Steps

Step 1: Register a Genesis Audio Device

1. Log in to GCXONE → Configuration App
2. Select your Site and navigate to the Devices tab
3. Click Add and set the type to Genesis Audio
4. Enter a name and click Discover
5. The system will generate unique SIP credentials — use these to configure your IP Speaker or SIP Horn

Step 2: Activate the Conference Toggle

1. In the Site tree view, select the newly added Genesis Audio device

2. Toggle the Active switch to ON
3. Repeat for all devices you want to include in the conference bridge

Step 3: Initiate an Audio Call

1. Open the Video Viewer
2. Drag a sensor or camera from the tree view into a tile
3. Locate the Mic Icon — default state is red
4. Click the icon to initiate the call — the icon will turn blue once the conference is active

Important Notes

Permissions: Audio functionality is only visible to users with the REMOTE_TWO_WAY_AUDIO permission enabled in their role.

Local Mode: Unlike video recording, audio calls do not require the Local Mode service to be active — they are handled directly via the cloud SIP gateway.

Bandwidth: High-fidelity audio adds approximately 64 kbps per participant. Ensure your network has sufficient overhead for multi-party conferences.